



Speaker-specific Patterns of Phonetic Covariation in Korean Word-medial Stops and the Role of Phonological and Morphological Contexts

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Abstract

Individual speaker differences in speech have often been regarded as random, but growing evidence suggests that variation across speakers follows systematic patterns. Yet, it remains an open question whether such structured variation is universal across languages, spans multiple phonetic dimensions, and remains consistent across contexts. Korean word-medial stops provide a compelling test case, as these stops are subject to extensive contextual variability. This study examines patterns in individual speaker variation by analyzing the phonetic distributions and covariation patterns between word-medial stops in Seoul Korean in varying phonological and morphological contexts. The results show that while individual differences arise in phonetic realization of these stops, speakers maintain structured covariation that allows stop categories to remain distinct despite variability, providing further support for the phonetic uniformity account.

Index Terms: individual speaker differences, phonetic covariation, contextual variation, phonetic uniformity

1. Introduction

The phonetic properties of speech sounds vary substantially across speakers, and this variation often appears arbitrary. However, recent research has shown that speakers exhibit systematic covariation across different sounds in a language, enabling predictions about the realization of one sound based on another. For instance, [1] demonstrated that voice onset time (VOT) in American English stop consonants is highly correlated across places of articulation within individual speakers, suggesting that phonetic variation reflects structured production patterns, governed by what has been termed phonetic uniformity [2]. While previous studies provide compelling evidence for covariation, it remains unclear whether such structured variation generalizes to various languages, phonetic dimensions, and linguistic contexts. For example, prior research has focused on word-initial stops, where phonological contrasts tend to be more robust [3], yet greater variability is often observed in word-medial position [4].

To address these gaps, the present study investigates Korean word-medial stops. If speakers exhibit consistent covariation in the phonetic properties of stop categories across various phonological and morphological conditions, this will provide further support for the phonetic uniformity principle, extending it to a language without a typical voicing contrast and beyond word-initial position. Moreover, such findings would suggest that phonologically and morphologically conditioned phonetic variation follows predictable patterns, reinforcing the idea that speaker-specific but structured constraints govern phonetic realization.

1.1. Phonetic characterization of Korean stops

Korean stop consonants are characterized by a three-way contrast among plain (lax/lenis), tense (fortis), and aspirated stops. This contrast in the word-initial position is primarily realized through differences in VOT and fundamental frequency (F0) at the following vowel onset: plain stops show intermediate VOT and lower F0 on the following vowel, tense stops show shortest VOT and raise F0, and aspirated stops show longest VOT and raise F0 [3].

While previous studies have primarily focused on word and/or phrase-initial positions, a focus on word-medial position is crucial because this is where the great degree of contextual variability occurs. Unlike word-initial stops, which are often subject to domain-initial strengthening [3], word-medial stops often undergo lenition [5] and neutralization that challenge phonological contrasts [6]. Investigating how speakers maintain structured phonetic relationships in these contexts provides insights into how humans navigate variability in speech.

Korean serves as an excellent test case since its word-medial stops exhibit great phonetic variability, particularly in intervocalic and morpheme boundary positions. This variability arises from intervocalic voicing of plain stops [7] and from compound tensification, where the initial plain obstruent of the second noun in a noun-noun compound surfaces as tense [8]. One key aspect of this variability is how phonetic properties like closure duration and F0 onset at the following vowel interact across contexts. Closure duration serves as a primary cue for distinguishing Korean stops in word-medial position [9], while F0 at the vowel onset remains a secondary cue more susceptible to prosodic effects [10]¹. Given these patterns, the present study investigates whether individual speakers maintain structured phonetic relationships across different phonological and morphological contexts.

1.2. Phonetic uniformity principle

A framework for understanding structured phonetic realization is phonetic uniformity [2]. This principle posits that speakers aim for consistency in their phonetic realizations across different contexts, even when phonetic pressures favor variation. Alongside factors such as auditory salience and articulatory ease [12], phonetic uniformity contributes to shaping structured relationships among related sounds. [1] provides evidence for this in American English, showing that individual speakers exhibit highly correlated voice onset time (VOT) values across different stop consonants. Their later work [13] extends this finding to sibilant fricatives, distinguishing between different

¹This is different from word-initial stops, where F0 at the onset of the following vowel has emerged as a primary cue due to ongoing tonogenesis in Seoul Korean [11].

types of uniformity. Similarly, [14] demonstrates covariation in Japanese stop voicing contrasts, with strong within-cue relationships and weaker between-cue correlations across speakers. Together, these findings suggest that phonetic uniformity operates as a constraint that gives rise to systematic, speaker-specific patterns of phonetic realization.

To further test this notion, the current production study examines whether the key phonetic properties for word-medial stops in Seoul Korean, closure duration and F0 onset of the following vowel, follow a structured realization across speakers and contexts in a controlled laboratory speech. If similar covariation patterns are observed, this would provide additional support that the uniformity principle applies across languages and phonetic dimensions, including F0. The results aligned with previous findings, showing that while individual speaker differences were evident across closure duration and F0 onset, the covariation patterns within these properties between stop categories remained systematic across speakers. However, the extent of this covariation differed by cue. Closure duration varied significantly with phonological and morphological context, whereas F0 onset appeared to be largely independent of these factors. This contrast raises further questions about how these distinct covariation patterns influence perception.

2. Methods

2.1. Stimuli

Participants read sentences containing 106 target words balanced for stop types (plain, tense, aspirated, nasal), place of articulation of the stops (POA; bilabial, alveolar), phonological contexts (post-vowel, post-sonorant, post-obstruent), and morphological contexts (simplex noun, compound). Each target word was embedded in a carrier sentence that included a name, a case marker, and a verb, resulting in sentences ranging from eight to ten syllables in length.

2.2. Data collection

Participants were 33 Seoul Korean L1 speakers with no reported speech or hearing disorders. Seventeen of these speakers (12 female, 5 male) were recorded in the U.S., while the rest (11 female, 5 male) were recorded in Korea. Compared to the speakers recorded in the U.S., those recorded in Korea did not have extensive experience living abroad. All speakers were between the ages of 19 and 39 and reported that they were raised in the Seoul and/or Gyeonggi area in Korea where the Seoul variety is spoken.

At the start of the experiment, participants reviewed the instructions with the experimenter and completed a set of practice trials. They were instructed to avoid pausing between words or phrases to maintain a natural speech rhythm. After the practice session, participants read sentences presented on the screen, with all materials and interactions conducted entirely in Korean. Each experimental block featured a unique, randomized set of words. The experiment consisted of four blocks, resulting in a total of 424 tokens per speaker.

2.3. Data processing

The dataset initially contained 13,992 tokens. After removing 424 tokens from one male speaker due to poor recording quality, 304 tokens for disfluencies or speech errors, 1,240 tokens for being in a morphological context other than simplex nouns and subordinate compounds, 12,024 tokens remained (85.9% of the

total). Forced alignment with Kaldi [15] was performed across speakers, followed by manual adjustments of all tokens.

For closure duration analysis, 366 tokens with no closure (e.g., spirantized plain stops) and 2,019 nasal stop tokens were excluded. Additionally, 1,022 tokens with unclear stop-stop boundaries and 25 outliers were removed, leaving 7,628 tokens (63.44%). F0 values were extracted from the vowels following word-medial stops using a cross-correlation pitch-tracking algorithm [16]. Extracted F0 values were removed if they exhibited abrupt jumps relative to adjacent values or appeared as isolated measurements (278 tokens). Additionally, 255 tokens with fewer than three F0 values over the vowels were excluded, as well as 80 outliers, leaving 11,477 observations (95.45%).

2.4. Analysis

2.4.1. Regression modeling of phonetic variation

To examine how closure duration and F0 onset vary across contexts, a Bayesian regression analysis was conducted in R [17] using the brms package [18]. Closure duration and F0 onset values were z-scored and analyzed separately as response variables. The models tested the effects of three main predictors: Stop Type (plain, tense, aspirated), Morphological Context (simplex, compound), and Preceding Context (non-high vowel, obstruent, sonorant). In addition to the main predictors, Speech Rate and Word-initial Type were included as covariates. Speech Rate was calculated as the inverse of the difference between the total duration of the utterance and the total duration of the stop, a measure designed to minimize correlations with other factors that could bias the coefficient estimates [19]. This measure was scaled by the minimum and maximum speech rate to range from 0 to 1. Word-initial Type refers to the type of sound at the beginning of a word, which influences the overall F0 level of the word [20]. Using default priors, model selection was conducted in a top-down approach, comparing various models with and without each factor from the maximally specified model. The final model included random effects, with a random slope for phonological context grouped by subject, and a random intercept for word item.²

2.4.2. Covariation patterns

Individual speakers' covariation in closure duration and F0 onset between stop pairs was analyzed to identify systematic speaker-specific patterns between stop categories. Covariation provides information about how changes in one phonetic variable influence another, offering a means to assess the relationship between the phonetic realization of two sounds. To control for speech rate effects, each response variable was fitted to a linear regression model with speech rate in *R*. Residual values from these models were then used to calculate individual speakers' means. Finally, Pearson correlation coefficients were computed using these mean values, separately for each contextual condition.

²The selected regression models are as follows:
 $Closure\ duration \sim StopType * PrecedingContext * MorphContext + StopPOA + SpeechRate + WordInitialType + (1 + PrecedingContext|Subject) + (1|Word)$;
 $F0\ onset \sim StopType * PrecedingContext * MorphContext + Gender + StopPOA + SpeechRate + WordInitialType + (1 + StopType|Subject) + (1|Word)$.

3. Results

3.1. Distribution of phonetic properties

3.1.1. Closure duration of the word-medial stop

Speakers exhibited a wide variety of closure duration ranges, with a speaker showing the widest range (163.56 ms) displaying values twice as long as a speaker with the narrowest range (80.7 ms). Notably, male speakers showed shorter closure durations compared to female speakers, despite no clear gender-based differences in speech rate.

Speakers maintained this pattern of range across phonological contexts. That is, speakers with short ranges exhibited shorter ranges in different contexts as well, suggesting that they operate within a consistent range of closure durations. Despite the differences in range, all speakers were influenced by both phonological and morphological contexts (Figure 1). In post-vowel context (V_V), speakers exhibited the longest mean closure durations for tense stops and the shortest for plain stops. However, in post-obstruent context (O_V), speakers showed similar distributions of closure durations, with plain stops often exceeding the durations of other stops. In post-sonorant (S_V) context, speakers preserved the distinction between stop types in simplex nouns, while these distinctions were neutralized in compounds.

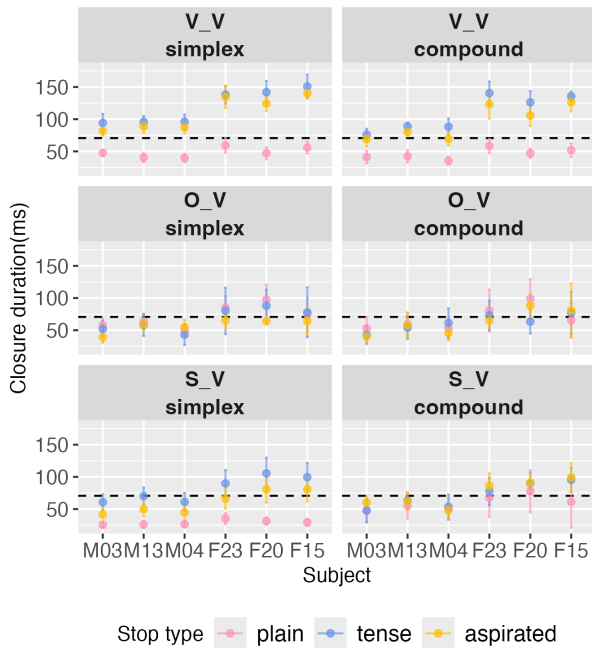


Figure 1: Distributions of closure duration by subject. The first three subjects show the narrowest range, and the last three subjects show the widest range. Rows indicate phonological contexts, and columns indicate morphological contexts. The dotted line shows the overall mean across speakers and conditions.

There was a clear subject-dependent pattern according to the varying subject coefficients ($\sigma(\text{Intercept})=0.37$, 95% CI=[0.29, 0.47]). Negative correlations were observed between the subject intercept and the effect of O_V ($\sigma(\text{Intercept}, \text{O_V})=-0.56$, 95% CI=[-0.77, -0.27]), and the subject intercept and the effect of S_V ($\sigma(\text{Intercept}, \text{S_V})=-0.69$, 95% CI=[-0.75, -0.44]). These indicate that speakers with shorter tense durations are

more sensitive to the effects of O_V and S_V. On the other hand, speakers with longer tense durations (i.e., higher intercepts) were more resistant to the effects of the contexts, suggesting that speakers with different baseline durations exhibit different sensitivity to the preceding context effect.

3.1.2. F0 onset of the following vowel

Speakers also exhibited a wide variety of F0 onset ranges. Similarly to closure duration, male speakers showed a narrower range (narrowest: 35.94 Hz) compared to female speakers (widest: 201.08 Hz). However, what is different from the patterns of closure duration is that F0 onset overlapped between stop categories across all contexts (Figure 2).

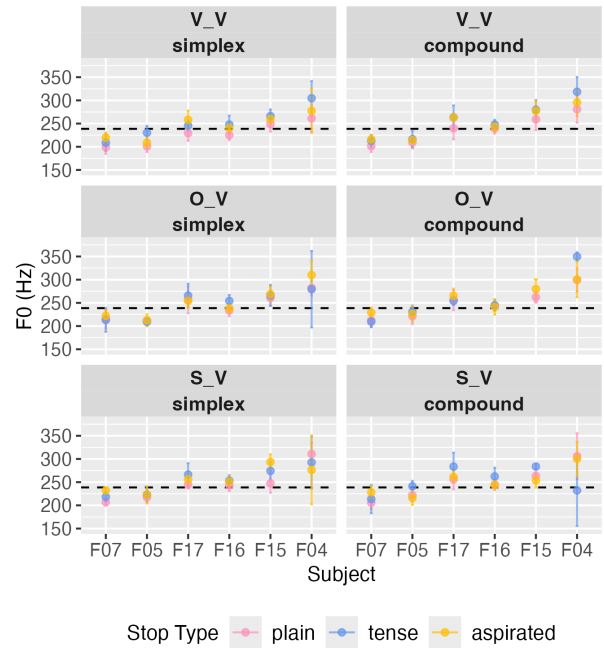


Figure 2: Distributions of F0 onset by subject. The first three subjects exhibit the narrowest range, while the last three subjects exhibit the widest range among female speakers. The format is consistent with Figure 1.

Nonetheless, the varying subject effects indicate substantial speaker-level variation, with this model's varying subject intercept exceeding that of the closure duration model's varying subject intercept ($\sigma(\text{Intercept})=0.47$, 95% CI=[0.36, 0.61]). The random slope effects for stop types showed slight variability, with the larger variation observed for aspirated stops ($\sigma(\text{Aspirated})=0.13$, 95% CI=[0.10, 0.18]) compared to plain stops ($\sigma(\text{Plain})=0.08$, 95% CI=[0.06, 0.11]). The correlation estimates between intercepts and stop types were weaker and less consistent ($r(\text{Intercept}, \text{Plain})=-0.26$, 95% CI=[-0.59, 0.11], $r(\text{Intercept}, \text{Aspirated})=-0.20$, 95% CI=[-0.53, 0.16], $r(\text{Plain}, \text{Aspirated})=0.09$, 95% CI=[-0.29, 0.45]), suggesting greater individual differences in the F0 onset realization irrespective of linguistic contexts. The greater variance in the F0 onset model and the weak correlations between intercepts and stop categories suggest that speaker-specific factors, independent of linguistic context, play a more prominent role in shaping F0 patterns than in shaping closure duration.

3.2. Covariation patterns

3.2.1. Closure duration of the word-medial stop

Across all preceding contexts, a strong linear relationship emerged between the individual speakers' mean closure durations of word-medial plain and tense stops (Figure 3). Similar correlations were also observed between plain and aspirated stops, as well as between tense and aspirated stops. These findings indicate that while individual speakers vary in their overall closure duration ranges, the relative durations of stop categories remain systematically constrained. That is, if a speaker's plain stop duration is relatively short, their tense and aspirated stop durations are also likely to be relatively short.

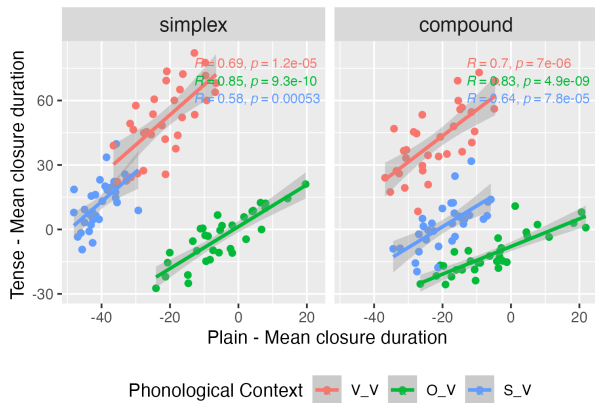


Figure 3: Covariation of subject-wise mean residualized closure durations for plain and tense stops.

Regarding contextual effects, speakers' mean durations are distinctly patterned by phonological contexts. For instance, post-obstruent plain and tense stops occupy a dimension clearly separable from post-sonorant plain and tense stops in Figure 3. However, the strength of this covariation pattern varies across contexts for all three stop pairs. For example, plain and tense stop durations show the strongest correlation in the post-obstruent context in simplex nouns ($r = 0.85$, $p < 0.001$), but plain and aspirated stops show the weakest correlation in the post-sonorant context in simplex nouns ($r = 0.49$, $p < 0.05$). Understanding the underlying mechanisms driving this contextual variation, as well as their perceptual implications for listeners, requires further investigation.

The effect of morphological context is less salient. Post-obstruent plain and tense stops in simplex nouns and compounds occupy a similar region, whereas post-sonorant plain and tense stops are clearly separable. However, this pattern aligns with expectations, given that post-obstruent tensification [21] influences the realization of plain stops in both morphological contexts, while post-sonorant plain stops in compounds are most likely to undergo compound tensification [8]. These findings suggest that while morphological context plays a role, its effects are more nuanced and largely shaped by the phonetic realization of (morpho)phonological processes.

3.2.2. F0 onset of the following vowel

Unlike closure duration, F0 onset exhibited an almost perfectly positive linear relationship across all contexts for vowels following all three pairs of stops. Figure 4 illustrates this relationship, showing that speakers' mean F0 onset values for vowels

following plain and tense stops largely overlap across contexts. This consistently linear yet overlapping distribution suggests that while F0 onset is constrained by speaker-specific patterns, it is only marginally influenced by phonological and morphological contexts. Whether this reflects the reliability of F0 cues remains to be investigated.

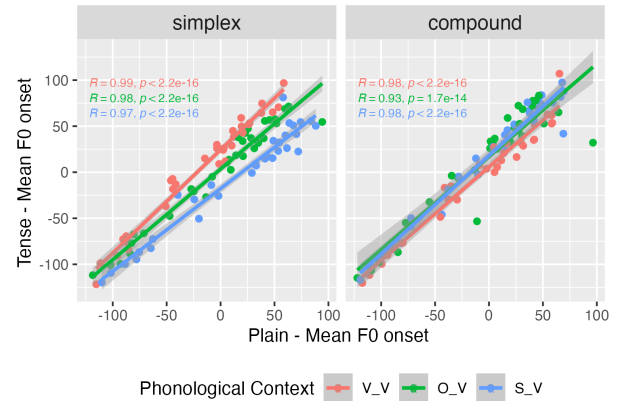


Figure 4: Covariation of subject-wise mean residualized F0 onset in vowels following plain and tense stops.

4. Discussion

The findings of this study demonstrate that phonetic variation in Korean word-medial stops follows a structured, speaker-specific pattern. While individual speakers consistently modulated closure duration based on phonological and morphological contexts, strong covariation patterns indicate that speakers maintain stable phonetic relationships between stops. In contrast, F0 onset patterns exhibited a different trend: although it showed strong speaker-specific covariation, it varied independently of phonological and morphological contexts. Further research is needed to determine whether this pattern reflects the reliability of F0 as a perceptual cue, particularly in light of the ongoing tonogenesis in Seoul Korean [11].

Altogether, this study offers insights into not only how individual speakers produce speech sounds in systematic patterns, but also how listeners may process and interpret them. The observed covariation within closure duration and F0 onset suggests that speakers maintain an internal phonetic structure that preserves relative relationships among stop categories, even as absolute values shift due to contextual factors. This structured covariation likely supports talker normalization, enabling listeners to efficiently adjust for speaker-specific variability [1]. The relationship between these covariation patterns and listeners' perception warrants further investigation.

Another promising direction for future work is to examine between-cue covariation, of how the coordination between multiple cues informs listeners' perception of phonemic contrasts, particularly when those contrasts are acoustically weak or ambiguous [6]. Investigating these patterns can reveal how listeners integrate information across cues to maintain perceptual stability, offering deeper insight into the mechanisms that support robust speech perception in the face of variability.

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